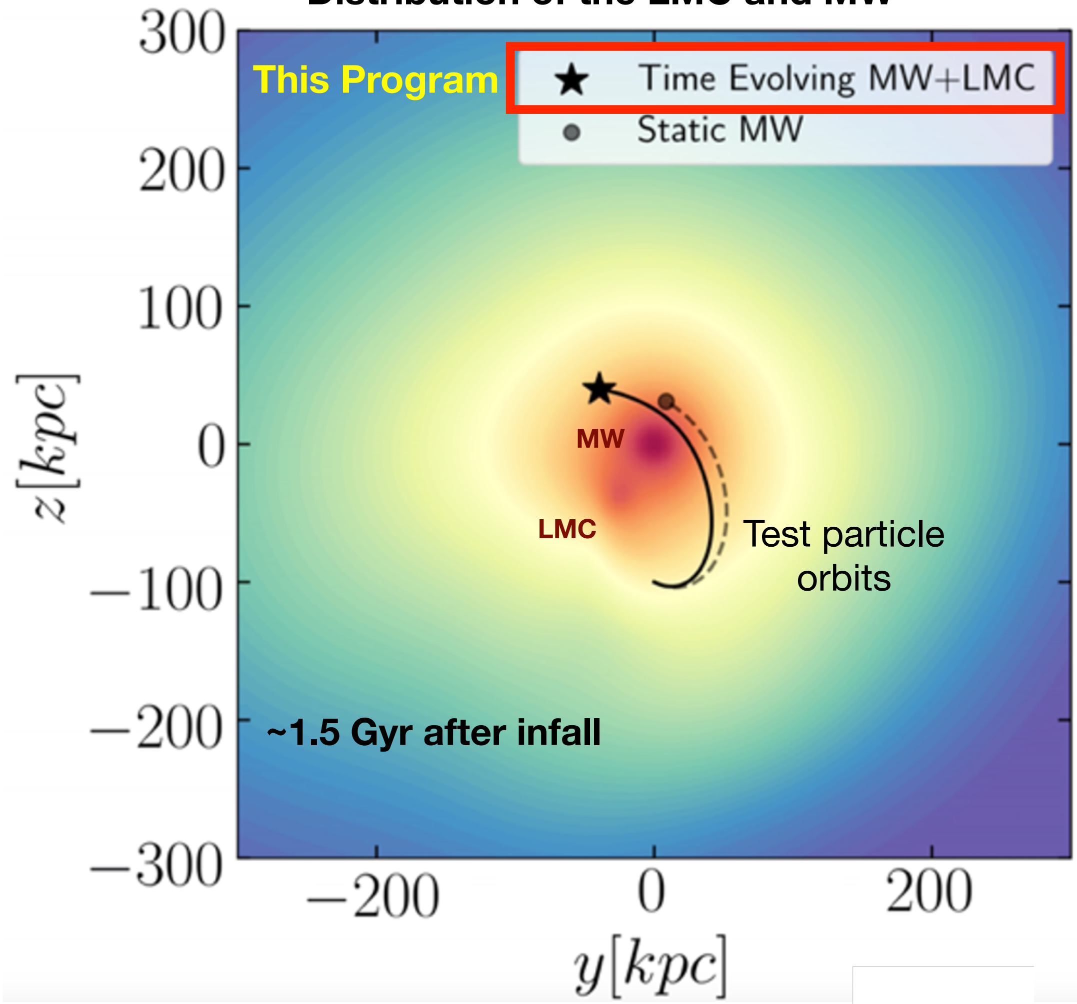
Figure & Caption

The motions of all objects in orbit about the Milky Way (MW) are impacted by the dark matter distortions caused by the recent infall of the LMC. The color scale illustrates the combined dark matter distribution of the MW and the LMC, 1.5 Gyr after the LMC's infall to the MW. The dashed line illustrates the orbit of a test particle, assuming a static MW. The solid line illustrates the orbit of the same test particle, but now accounting for the deformation of the MW's dark matter caused by the LMC. Ignoring the time evolving dark matter perturbations from the LMC will cause errors in orbital calculations. <u>This program will build high-resolution N-body simulations to quantify the time evolving MW+LMC dark matter potential.</u> Combined with HST/Gaia astrometry, these new potentials will enable the orbit reconstruction of *all* MW halo tracers (globular clusters, satellites, streams) necessary to constrain the dark matter distribution of our Galaxy.

The Combined Cold Dark Matter Distribution of the LMC and MW



NOTE: there should have been a color bar